

Virtualization Technology and Cloud Computing Lab

Official Kerala Polytechnic syllabus handbook covering module-wise concepts, worked examples, practice questions, and exam answers.

Course Code

6139A

Credits

4

Periods

5/week

Course Details

Title: Virtualization Technology and Cloud Computing Lab

Department: Common

Revision: REV_Virtualization Technology and Cloud Computing Lab

Pattern: Theory / Practical

Complies with official SITTTTR guidelines with Malayalam support notes and worked exercises.

Curriculum integrity

Official Source Declaration

This handbook's course identity is aligned with the Revision 2026 master subject index for **Course 6139A — Virtualization Technology and Cloud Computing Lab**. Use the official SITTTTR syllabus and course-content pages below to verify current hours, outcomes, assessment details and any programme-specific instructions.

[Official SITTTTR Revision 2026 syllabus reference](#). This page is a study handbook, not a substitute for the latest official notification or examination instruction.

Malayalam support: ു ു൬൬൬൬൬൬ ു൬൬൬൬൬൬൬ ു൬൬൬൬൬൬൬ English technical terms- ു൬൬൬൬൬൬൬൬൬. ു൬ ു൬൬൬൬൬ ു൬൬൬൬൬൬൬ ു൬൬൬൬൬൬൬, formula, diagram, units, definitions ു൬൬൬ ു൬൬൬൬൬൬൬ ു൬൬൬൬൬൬൬ ു൬൬൬൬.

Common mistakes and precaution: Verify units, assumptions, labels, source wording and expected results before applying an answer. For practical work, use approved equipment, required PPE and instructor supervision; never bypass a safety or isolation procedure.

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OBJECTIVES

Course Objectives and Outcomes

Objectives

1. Provide deep technical understanding of Virtualization Technology and Cloud Computing Lab.
2. Develop practical competencies aligned with industry standards.
3. Prepare students for board examinations.

CO-PO Mapping

CO1 → PO1: 3. CO2 → PO2: 3. CO3 → PO3: 3. CO4 → PO3: 3.

MODULE 1

Core Principles

Outcomes

Master foundational terminology and core definitions of Virtualization Technology and Cloud Computing Lab.

Study

Detailed analysis of core principles and theoretical foundations.

Malayalam Support Note: വിദ്യാർത്ഥികൾക്ക് വിഷയത്തിന്റെ അടിസ്ഥാനപദങ്ങളും ആവശ്യമായ പദപരിഭാഷയും നൽകുന്നതിനായി ഈ കുറിപ്പ് ഉൾക്കൊള്ളുന്നു.

MODULE 2

Applied Methods

Outcomes

Apply standard calculation techniques and operational procedures.

Study

Numerical problem-solving techniques and practical workflows.

Malayalam Support Note: ഐക്യരൂപം ഉപയോഗിച്ച് സമവാക്യം പരിഹരിക്കുക.
 ഐക്യരൂപം ഉപയോഗിച്ച് പരിഹരിക്കുക.

MODULE 3

Advanced Integration

Outcomes

Evaluate complex systems and optimize performance.

Study

Advanced system integration and troubleshooting methodologies.

Malayalam Support Note: സങ്കീർണ്ണമായ സിസ്റ്റം ഇൻ്റഗ്രേഷൻ രീതികൾ പഠിക്കുക.
 സങ്കീർണ്ണമായ സിസ്റ്റം ഇൻ്റഗ്രേഷൻ രീതികൾ പരിഹരിക്കുക.

MODULE 4

Synthesis & Case Studies

Outcomes

Design comprehensive technical solutions and demonstrate mastery.

Study

Real-world case studies and industry project design.

Malayalam Support Note: ഐ.ഐ.ടി. മലയാള സാങ്കേതികവിദ്യയിൽ പ്രവർത്തിക്കുന്നവർക്ക്
ഐ.ഐ.ടി.യുടെ സാങ്കേതികവിദ്യയെക്കുറിച്ച്.

MODEL PAPER

Practice Model Question Paper

Time: 3 Hours | **Max Marks:** 75

Part A (2 marks each)

1. Define core concepts of Virtualization Technology and Cloud Computing Lab.
2. Explain fundamental principles in Module 1.
3. Discuss calculation methods in Module 2.
4. Describe advanced integration in Module 3.
5. Summarize case studies in Module 4.

Part B (5 marks each)

1. Detailed explanation of Module 1 concepts with examples.
2. Comprehensive analysis of Module 2 methods.
3. System integration and troubleshooting in Module 3.
4. Industry applications and design in Module 4.

ANSWERS

Full Answer Key

Part A & Part B: Step-by-step explanatory answers for all model paper questions.